

LÉHO DEHOVE

R&D ENGINEERING INTERN

BIO-MECHATRONICS AND ROBOTICS SPECIALIZATION

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PROFILE

Final-year Bio-Mechatronics Engineering student with hands-on experience in prototyping, machine learning, CAD, additive manufacturing, and electronics applied to robotics. Combines control theory, kinematics, and low-level firmware with neuroscience research (CNRS / ESPCI PSL) and industrial manufacturing (Danfoss). Seeking a 6-month R&D internship from April 2027.

SKILLS

Robotics, Vision & Control

ROS / ROS2, OpenCV, Linux
Orocos / BFL, MATLAB/Simulink
Kinematics, Dynamics, SLAM

Embedded & Electronics

C, C++, Python, Rust, Git
STM32, PIC16F1619, ESP32, Arduino
MPLAB X / MCC Melody, I2C/SPI, PCB

Prototyping & CAD/CAE

SolidWorks, CATIA, Abaqus/CAE, Rhino
3D Printing: 10+ years (FDM/SLA/SLS)
CNC Milling, Laser, Wood/Metal

ENGINEERING EXPERIENCE

CNRS – ESPCI PSL University (Brain Plasticity Lab)

June – expected Sept. 2026

Computer Vision R&D Intern

- Developing video processing software for automated kinematic analysis in animal behavior research, increasing throughput by 500%, saving over 45 hours per batch, and turning a week-long task into a single day.
- Co-authoring a research article on automated neural locomotion kinematics under optical stimuli.
- Designed and upgraded an optoelectronic hardware module for phototactic recording behavior.

Danfoss Power Solutions II (Manufacturing Plant)

June – July 2025

Manufacturing & Quality Operator

- Packaging, quality inspection, and assembly of small and medium-sized mechanical components for fluid power systems.

Enedis Business France (Energy Management Division)

July – Aug. 2023

Operations & Project Management Intern

- Supported national rollout of Linky smart meters through cross-team coordination, structured project tracking, and operational reporting.

ESME FabLab & Robotics Club

Sept. 2022 – June 2025

FabLab Administrator & Club Vice-President

- Managed digital fabrication machines (CNC, SLA/FDM 3D printing, laser cutters) and trained 150+ engineering students in CAD/CAM and rapid prototyping.
- Coordinated student club activities, cross-functional engineering projects, and robotics events.

PROJECTS

3D Particle Manipulation Platform | Custom Power Electronics · ROS · Microcontrollers

2025 – 2026

- Custom power electronics and ROS to control dust particles in a 3D enclosed space. Programmed deterministic control firmware.

Hand Exoskeleton for Rehabilitation | Biomechanics · Motorized Actuation · Embedded Control

2025

- Motorized hand exoskeleton to assist with daily muscle weakness and physical rehabilitation (functional physical prototype).

Autonomous Ball-Tracking & Mobile Robot | ROS · OpenCV · Autonomous Navigation

2024 – 2025

- Autonomous robot capable of navigating, detecting, catching, and throwing a ball. Winner of the robotics tournament.

Upper Limb Prosthetic Device | Biomechanics · Ergonomics · Material Physics · Additive Mfg

2023 – 2024

- Custom mechanical prosthetic and orthotic device developed for a real child with upper limb agenesis, focusing on ergonomics, material physics, and biomechanical fit without electronics.

EDUCATION

Master of Engineering – Bio-Mechatronics Specialization

Expected Sept. 2027

ESME Engineering School (Mechanical, Electrical & Electronics Curriculum)

Ongoing Key Courses : Medical Robotics, Biomechanics, Ergonomics for Design, Modeling and Simulation, Impedance Control, Robust Control, Biomaterials, Computer Vision, Real-Time Computing.

Exchange Student – Electrical & Electronic Engineering

2025 – 2026

Universiti Putra Malaysia (UPM)

IEEE final year collaboration across international cultures | English Proficiency: C1 Academic English (IELTS 6.5 / TOEIC 875).

ADDITIONAL INFORMATION

- Languages:** French (native), English C1 (IELTS 6.5 / TOEIC 875), Spanish B1.
- Interests:** Music (piano & guitar), Sports (weightlifting, climbing, seasonal), Video & board games (strategy), Raspberry Pi side projects, Science communication, 10+ years additive manufacturing.
- Licenses:** Driving License (Category B and AM), Coastal Boating License (Permis Côtier).